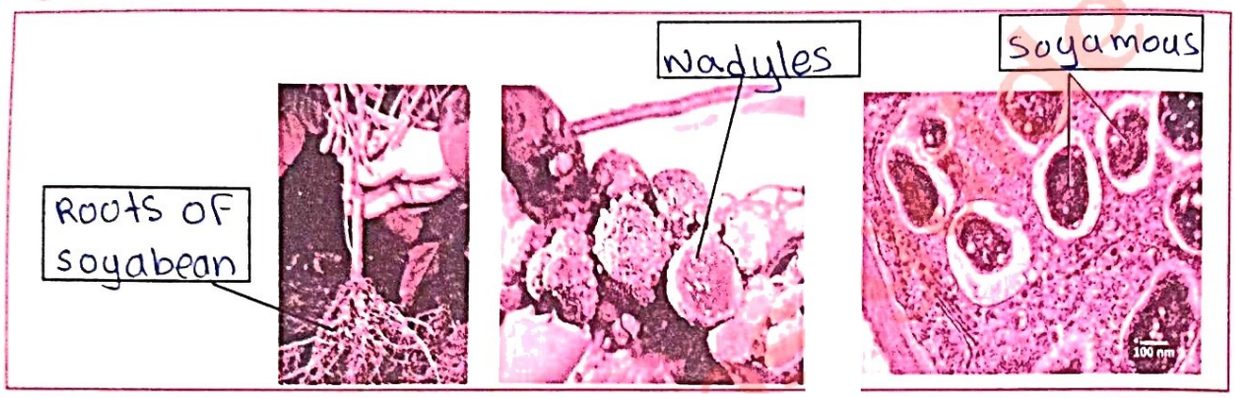
Experiment No.6

Title : Study of bacteria present in root nodules of leguminous plants.

Apparatus/ Materials : Rooted plantlets of either fenugreek/peanuts/sweet pea or beans, glass slide, cover glass, dropper, compound microscope, new razor, etc.

Chemicals : Safranin, 3% – 5% hydrogen peroxide (H_2O_2) solution, 70% ethyl alcohol (C_2H_5OH), etc.

Figure :



Procedure :

1. Take the rooted plantlet of any one of the above mentioned plant & sterilize its roots with the help of 3% – 5% hydrogen peroxide (H_2O_2) solution.
2. Deep the roots in 70% ethyl alcohol (C_2H_5OH) for 4 – 5 minutes.
3. Wash the roots with sterile water & take the thin sections of root nodules with the help of new razor blade.
4. Select a good section & keep it in safranin for 2 – 3 minutes.
5. Keep that section on glass slide, cover it cover glass & observe under 45X objective of compound microscope.

Observation : Rhizobium bacteria appear like pink, rod-like. The pinkish shape rhizobium bacteria are nutieed under compound microscope.

Inferance

Rhizobium bacteria present in root nodules of leguminous plants help to increase the protein content of pulses by the process of nitrogen fixation. And in exchange in the form of carbondrate from it such a mathual benifical relationship is called symbols and this bacteria

Multiple Choice Questions

1. Which stain is used for observation of rhizobia?
   A. Safranin ✓ B. Hydrogen peroxide C. Ethyl alcohol D. Eosin
2. According to shape, rhizobia are which type of bacteria?
   A. Cocci. B. Bacilli. ✓ C. Spirilli. D. Clustered bacilli.
3. Rhizobia are absent in.....
   A. fenugreek. B. pea nut. C. soy bean. D. millet (bajra). ✓

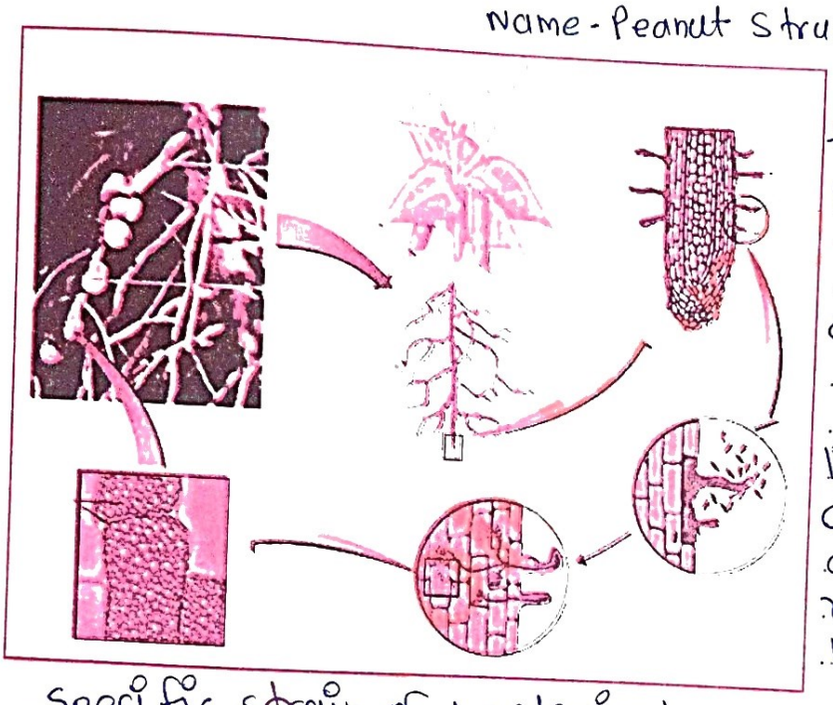
4. Due to rhizobia, ...content of beans & pulses increases.
A carbohydrates B. vitamins C. proteins D. minerals

5. Why is it necessary to sterilize the roots of rhizobia in this experiment?
A. To prevent the destruction of rhizobia.
B. To prevent the bacterial infection to us.
C. To prevent the mixing of other bacteria with rhizobia.
D. None of these.

Questions

1. Give the names in following diagram & explain it..

Name - Peanut Structure.



Root nodules are found on the root of plants, primary legumes, that form symbiosis with nitrogen fixation.

under nitrogen limiting condition, capable plants form a symbiotic relationship with host. ~~spo~~

specific strain of bacteria known as rhizobia.

2. What is symbiotic nutrition (commensalism)?

Rhizobia living in root hoduces supply nitrates and aminoacid to that plant and in exchange get the energy inform of carbohydrate from it such a matually benifical rodiation is called symbition.

3. Root nodules are not present on the roots of all the plants. How does the nitrogen fixation occur in those plants which do not have root nodules?

Rizobium bacteria converts atmosphic hitrogen into sulphides of nitrogen that plants an use.

H6EDLA